

## International Marketing Analytics

# Forecasting and time series

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WT 2026/27

# Today

Concepts | By hand | How good? | Real data | Python | In class

## Theory, Parts 1 to 3

- Trend, season, cycle; stationary or not?
- Simple forecasts and smoothing by hand
- How good is a forecast?

## Practice, Parts 4 to 6

- Real data: clothing sales in Australia
- Python: a code ladder in ten steps
- Online quiz and its answers
- Coding exercise and Posit Assistant

**Question of the day:** how much will we sell next month, and how sure can we be?

# Our example: planning demand in two shops

**The problem:** a marketing manager must plan before demand arrives.

**The question:** how much will we sell next week, next quarter, next Christmas?

## Stock

order before the peak

## Staff

shifts for the busy weeks

## Media budget

spend when buyers are there

**Two tiny shops first:** a phone shop (contracts per week) and a chocolate web shop (sales per quarter); real data in Part 4.

Invented round numbers (hand_examples.csv), built for this course

# The data: one row per period

## Phone shop: new contracts per week

| Week | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|---|
| **Contracts** | 40 | 44 | 42 | 46 | 50 | 49 | 53 | 54 |

## Chocolate shop: sales per quarter, k EUR

| | Q1 | Q2 | Q3 | Q4 |
|---|---|---|---|---|
| **Year 1** | 20 | 16 | 20 | 30 |
| **Year 2** | 24 | 20 | 23 | 35 |
| **Year 3** | 28 | 24 | 26 | 40 |

**period:** the time step: week 1 to 8, quarter 1 to 12

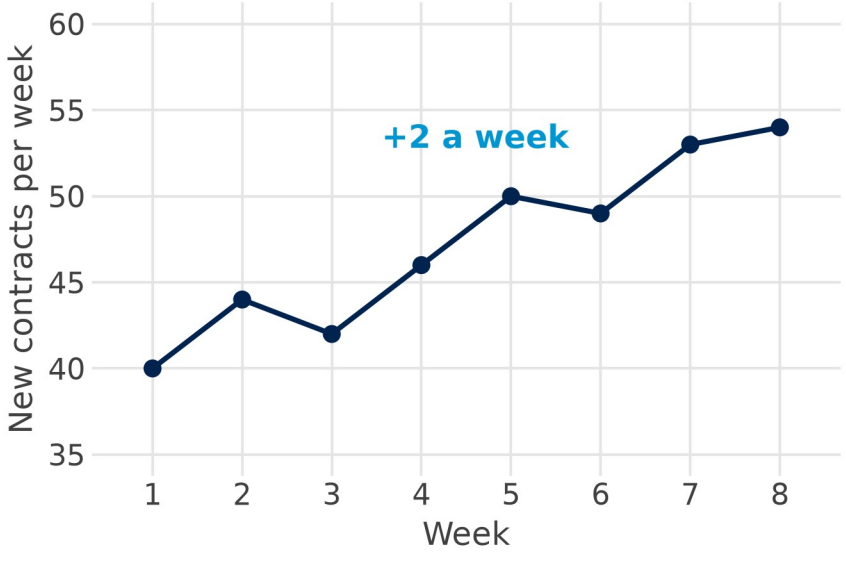
**value:** what we forecast

**rules:** time order, equal steps, no gaps

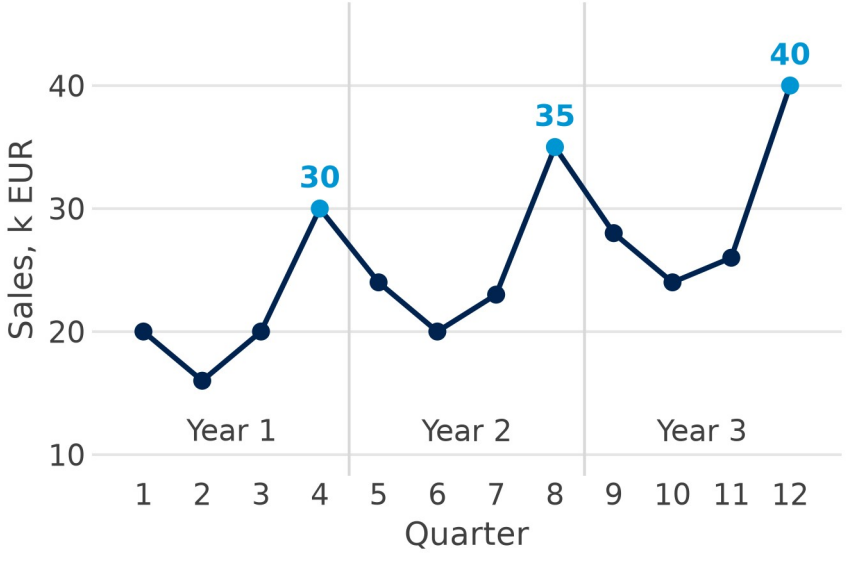
**File:** `data/forecasting/hand_examples.csv` invented, 20 rows: example, period, value

Invented round numbers (hand_examples.csv)

# First look: plot the series




Trend: up about 2 a week




Trend and season: every Q4 peaks

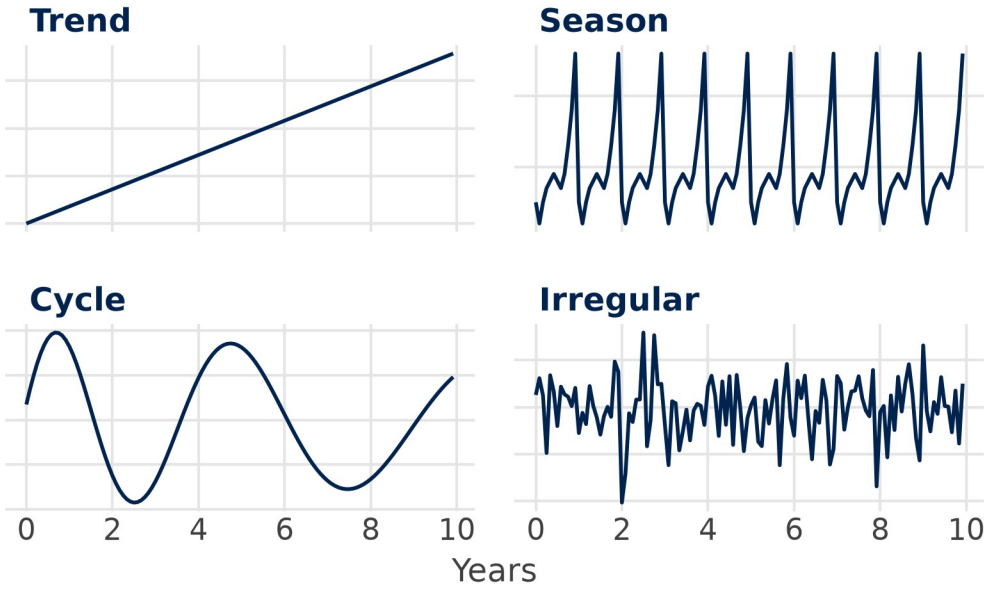
**Plot first:** the eye finds trend and season before any model.

Invented round numbers (hand_examples.csv)

# Time series concepts

Part 1 of 6 · trend, season, cycle, stationarity

# Trend, season, cycle and irregular




**Trend:** the long-run direction

**Season:** repeats every year, fixed length (Christmas)

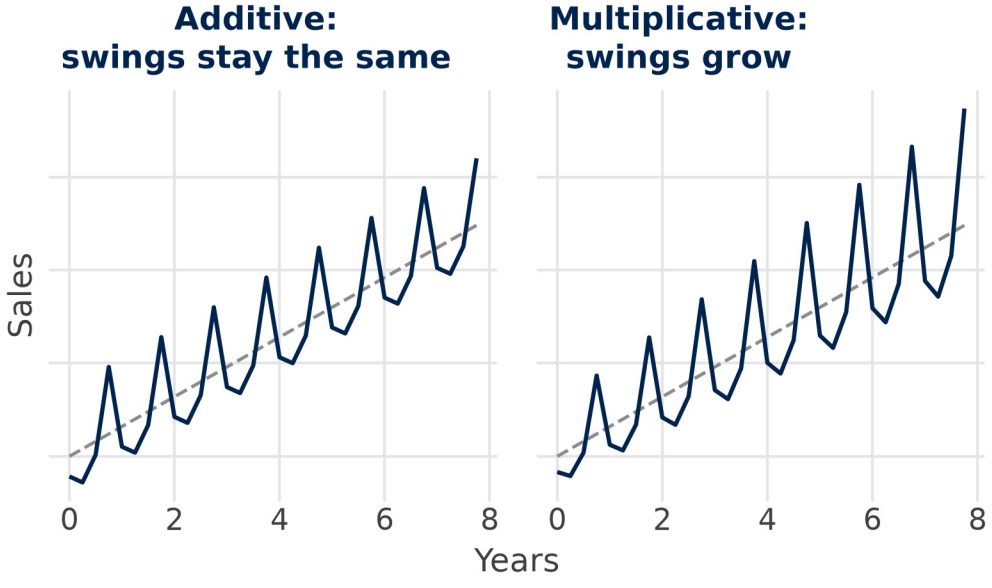
**Cycle:** ups and downs over years, no fixed length (the economy)

**Irregular:** what is left: noise, one-off events

**Series = trend + season + cycle + irregular (or ×, slide 8):** a forecast extends the first three.

Simulated series for illustration

# Additive or multiplicative season?




**+9.5**

**chocolate shop:** Q4 about 9.5 k EUR above the year mean on average (8.5 to 10.5): additive

**× 1.59**

**clothing (Part 4):** December 59 % above an average month, and the swing grows: multiplicative

**Multiplicative season:** use a multiplicative model or take logs, which makes it additive.

Simulated series for illustration; chocolate shop: invented; clothing: ABS Retail Trade (8501.0), CC BY 4.0

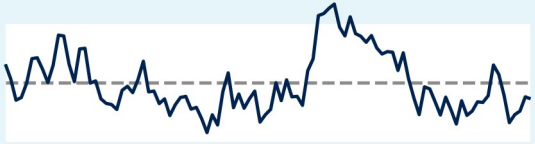
# What does stationary mean?

**Stationary:** the level and the ups and downs stay the same over time; any stretch of the series looks like any other stretch.

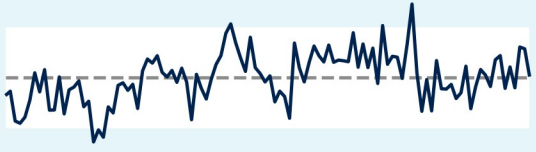
**Stationary**
noise around a fixed mean



**Stationary**
wobbles, returns to its mean



**Stationary**
clothing: change on last year



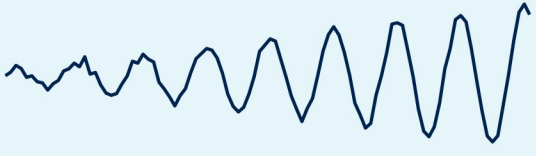
**Not stationary**
a trend: the level rises



**Not stationary**
a random walk wanders off



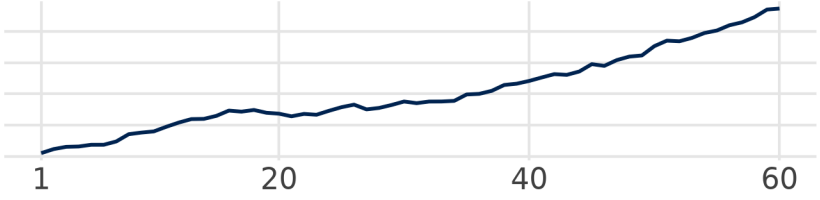
**Not stationary**
a season whose swings grow



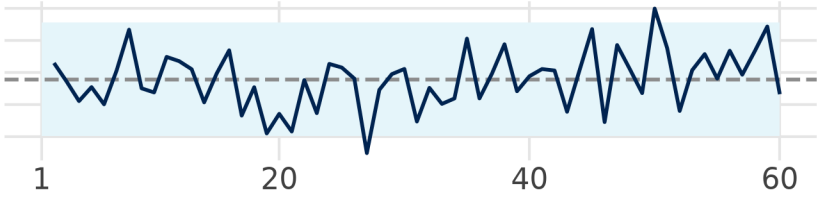
Simulated series for illustration; clothing: ABS Retail Trade (8501.0), CC BY 4.0

# Not stationary? Differencing is the fix

**A random walk with drift: it rises and wanders**




**Its changes (this period − the last): stationary**




Period

| Week | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | Average |
|---|---|---|---|---|---|---|---|---|---|
| **Contracts** | 40 | 44 | 42 | 46 | 50 | 49 | 53 | 54 | |
| **Change** | – | +4 | −2 | +4 | +4 | −1 | +4 | +1 | +2 |

**Differencing:** this period − the last; for a season: minus the same period last year. ARIMA-type models need it.

Simulated series for illustration; phone shop: invented round numbers

# Simple forecasts by hand

Part 2 of 6 · five methods and smoothing

# No change and same change: naive and drift

| Week | Contracts | Naive | Error = change |
|---|---|---|---|
| **1** | 40 | – | – |
| **2** | 44 | 40 | +4 |
| **3** | 42 | 44 | −2 |
| **4** | 46 | 42 | +4 |
| **5** | 50 | 46 | +4 |
| **6** | 49 | 50 | −1 |
| **7** | 53 | 49 | +4 |
| **8** | 54 | 53 | +1 |

**54**

**naive for week 9:**
= week 8

**56**

**drift for week 9:**
54 + the mean error
(54 − 40) / 7 = +2

**Naive's errors are the weekly changes;** drift adds their average, +2.

Invented round numbers (hand_examples.csv)

# Averages: the mean and a moving average

| Week | Contracts | Calculation | Forecast | Error |
|---|---|---|---|---|
| 1 | 40 | – | – | – |
| 2 | 44 | – | – | – |
| 3 | 42 | – | – | – |
| 4 | 46 | (40 + 44 + 42) / 3 | 42 | +4 |
| 5 | 50 | (44 + 42 + 46) / 3 | 44 | +6 |
| 6 | 49 | (42 + 46 + 50) / 3 | 46 | +3 |
| 7 | 53 | (46 + 50 + 49) / 3 | 48.33 | +4.67 |
| 8 | 54 | (50 + 49 + 53) / 3 | 50.67 | +3.33 |

**47.25** **mean:** 378 / 8

**52** **week 9:** moving average of weeks 6 to 8

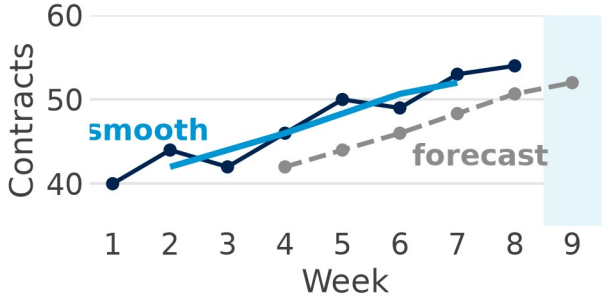
Sum: 40 + 44 + 42 + 46 + 50 + 49 + 53 + 54 = 378

**Averages lag behind a trend:** every error of the moving average is positive, the forecast is too low.

Invented round numbers (hand_examples.csv)

# Rolling averages: smooth or forecast?

| Week | Contracts | Centred window | Smooth | Forecast |
|---|---|---|---|---|
| **1** | 40 | – | – | – |
| **2** | 44 | (40 + 44 + 42) / 3 | **42** | – |
| **3** | 42 | (44 + 42 + 46) / 3 | 44 | – |
| **4** | 46 | (42 + 46 + 50) / 3 | 46 | **42** |
| **5** | 50 | (46 + 50 + 49) / 3 | 48.33 | 44 |
| **6** | 49 | (50 + 49 + 53) / 3 | 50.67 | 46 |
| **7** | 53 | (49 + 53 + 54) / 3 | 52 | 48.33 |
| **8** | 54 | – | – | 50.67 |
| **9** | ? | – | – | 52 |




**Smooth:** the week before, the week and the week after

**Forecast:** only the 3 weeks before (slide 13)

**Same averages, different place:** centre the window to see the trend; look only back to forecast.

Invented round numbers (hand_examples.csv)

# Exponential smoothing, step by step

| Week | Contracts | Level | Calculation (α 0.5) |
|---|---|---|---|
| **1** | 40 | 40 | start: the first value |
| **2** | 44 | 42 | 0.5 × 44 + 0.5 × 40 |
| **3** | 42 | 42 | 0.5 × 42 + 0.5 × 42 |
| **4** | 46 | 44 | 0.5 × 46 + 0.5 × 42 |
| **5** | 50 | 47 | 0.5 × 50 + 0.5 × 44 |
| **6** | 49 | 48 | 0.5 × 49 + 0.5 × 47 |
| **7** | 53 | 50.5 | 0.5 × 53 + 0.5 × 48 |
| **8** | **54** | **52.25** | **0.5 × 54 + 0.5 × 50.5** |

**Rule**

```
new level = α × value
          + (1 − α) × old level
forecast = last level
```

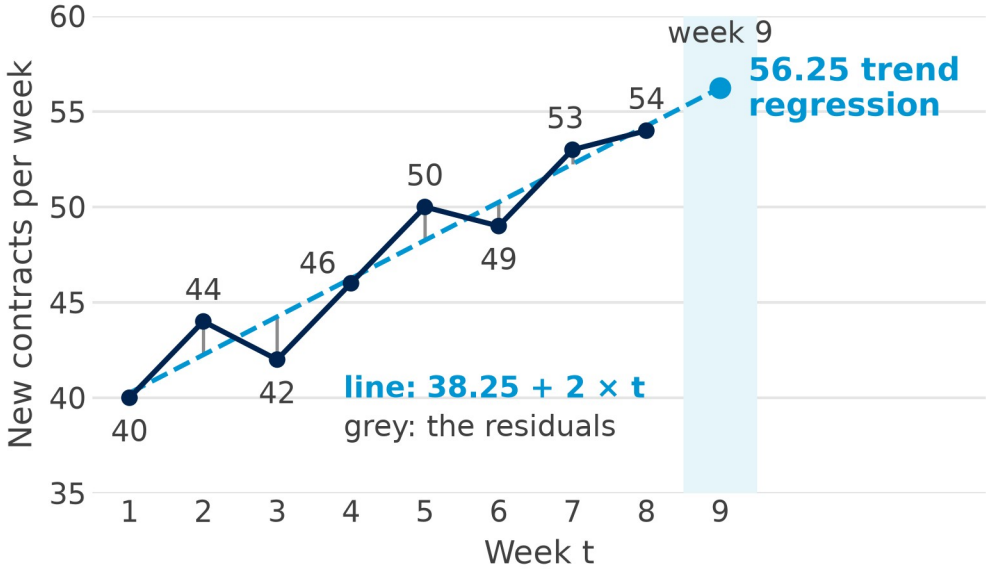
**52.25** **forecast** for week 9 (α 0.5)

**47.89** **α 0.2:** slower
α 1 = naive (54)

**Recent weeks count most:** α near 1 follows the data fast, α near 0 averages.

Invented round numbers (hand_examples.csv)

# Regression on time: the trend line




**By hand**

$\bar{t} = 4.5 \quad \bar{y} = 47.25$

$b1 = \Sigma(t - \bar{t})(y - \bar{y}) \,/\, \Sigma(t - \bar{t})^2$

$= 84 \,/\, 42 = 2$

$b0 = \bar{y} - b1 \times \bar{t}$

$= 47.25 - 2 \times 4.5 = 38.25$

week 9: $38.25 + 2 \times 9 = 56.25$

**0.926**

$R^2 = 1 - 13.5 / 181.5$

the line fits well

**The trend line uses every week:** it assumes the trend goes on – and that the errors are independent (slide 33).

Invented round numbers (hand_examples.csv)

# The trend line by hand, week by week

| Week t | Contracts y | $t - \bar{t}$ | $y - \bar{y}$ | $(t - \bar{t})(y - \bar{y})$ | $(t - \bar{t})^2$ | Fitted | Residual |
|---|---|---|---|---|---|---|---|
| **1** | 40 | −3.5 | −7.25 | +25.375 | 12.25 | 40.25 | −0.25 |
| **2** | 44 | −2.5 | −3.25 | +8.125 | 6.25 | 42.25 | +1.75 |
| **3** | 42 | −1.5 | −5.25 | +7.875 | 2.25 | 44.25 | −2.25 |
| **4** | 46 | −0.5 | −1.25 | +0.625 | 0.25 | 46.25 | −0.25 |
| **5** | 50 | +0.5 | +2.75 | +1.375 | 0.25 | 48.25 | +1.75 |
| **6** | 49 | +1.5 | +1.75 | +2.625 | 2.25 | 50.25 | −1.25 |
| **7** | 53 | +2.5 | +5.75 | +14.375 | 6.25 | 52.25 | +0.75 |
| **8** | 54 | +3.5 | +6.75 | +23.625 | 12.25 | 54.25 | −0.25 |
| **Sum** | **378** | **0** | **0** | **84** | **42** | **378** | **0** |

Means: $\bar{t} = 4.5$, $\bar{y} = 47.25$. Fitted = 38.25 + 2 × t; residual = y − fitted.

**b1 = 84 / 42 = 2, b0 = 47.25 − 2 × 4.5 = 38.25:** two sums give the trend line.

Invented round numbers (hand_examples.csv)

# Season by hand: the chocolate shop

| Sales, k EUR | Q1 | Q2 | Q3 | Q4 | Year mean |
|---|---|---|---|---|---|
| **Year 1** | 20 | 16 | 20 | **30** | 21.5 |
| **Year 2** | 24 | 20 | 23 | **35** | 25.5 |
| **Year 3** | 28 | 24 | 26 | **40** | 29.5 |
| **Season: average gap** | **−1.5** | **−5.5** | **−2.5** | **+9.5** | **sum 0** |

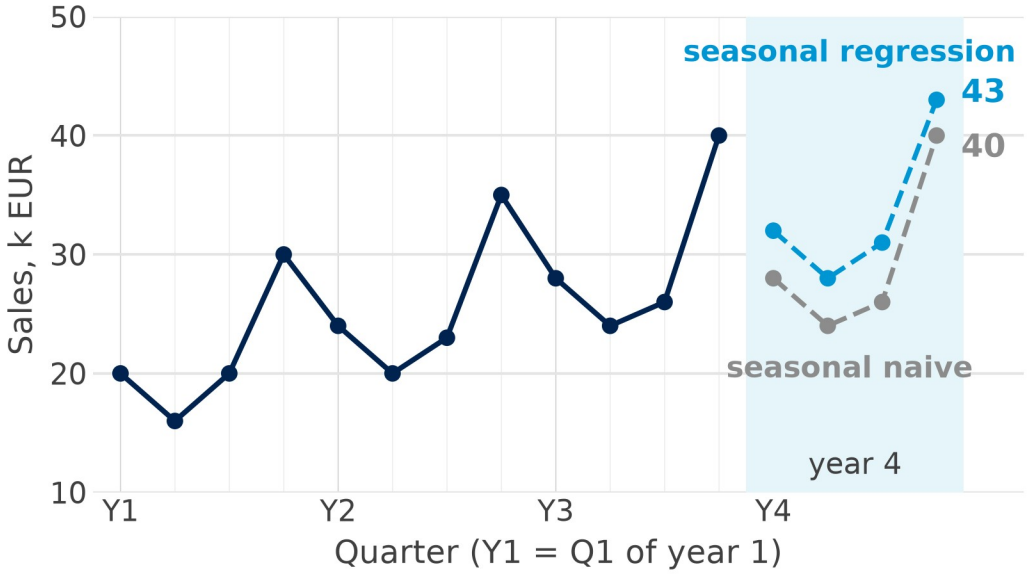
**Q4:** gaps 30 − 21.5 = +8.5, then +9.5, +10.5
average (8.5 + 9.5 + 10.5) / 3 = +9.5

**Trend:** year means 21.5 → 25.5 → 29.5
+4 a year, about +1 a quarter (slide 19)

**Season = the average gap to the year mean:** Q4 sells 9.5 k EUR more than an average quarter.

Invented round numbers (hand_examples.csv)

# Two seasonal forecasts for year 4




**40** **seasonal naive:**
Q4 of year 4
= Q4 of year 3

**43** **seasonal regression:**
$19 + 1 \times 16 + 8$
the trend at t = 16 plus the Q4 effect

Regression, $R^2$ 0.992:
$19 + 1 \times t - 5\ Q2 - 3\ Q3 + 8\ Q4$
Q2, Q3, Q4 = 1 in that quarter, else 0
(Q1 = base)

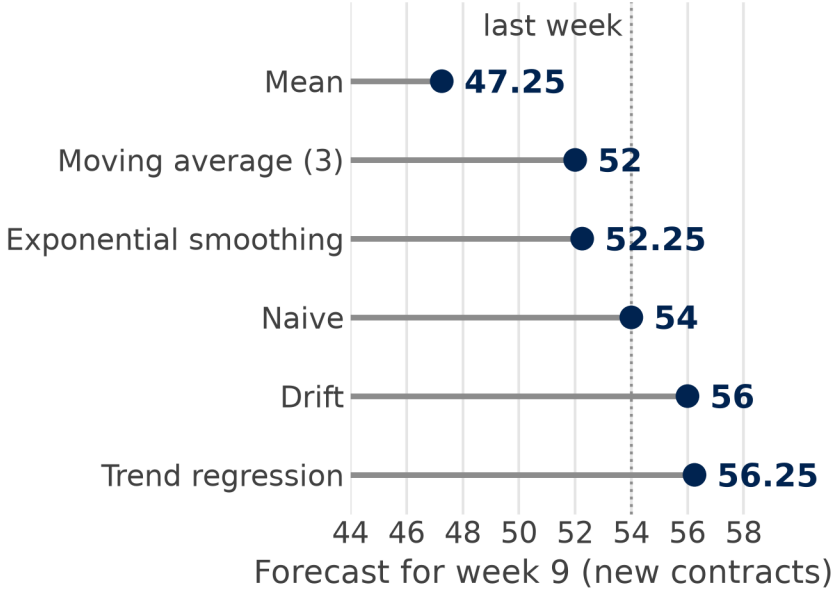
**Seasonal naive repeats last year;** the regression adds the trend of +1 a quarter.

Invented round numbers (hand_examples.csv)

# How good is a forecast?

Part 3 of 6 · hold out, measure, compare

# Six forecasts for week 9: which to trust?




| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|
| 40 | 44 | 42 | 46 | 50 | 49 | 53 | 54 |

**training period** **holdout: weeks 6–8**

**Step 1:** fit each method on weeks 1–5

**Step 2:** forecast weeks 6–8, compare with 49, 53, 54

**Never at random:** test weeks come last.

**Six methods, six answers, 9 contracts apart:** test each on weeks it has not seen.

Invented round numbers (hand_examples.csv)

# Errors and three measures, by hand

| Week | Actual | Naive | Error | \|Error\| | Error² | \|Error\| / actual |
|---|---|---|---|---|---|---|
| **Week 6** | 49 | 50 | −1 | 1 | 1 | 2.0 % |
| **Week 7** | 53 | 50 | +3 | 3 | 9 | 5.7 % |
| **Week 8** | 54 | 50 | +4 | 4 | 16 | 7.4 % |
| **Mean** | | | | **2.67** | **8.67** | **5.0 %** |

Naive from week 5: 50 for weeks 6, 7 and 8

**2.67** **MAE:** mean of |error|

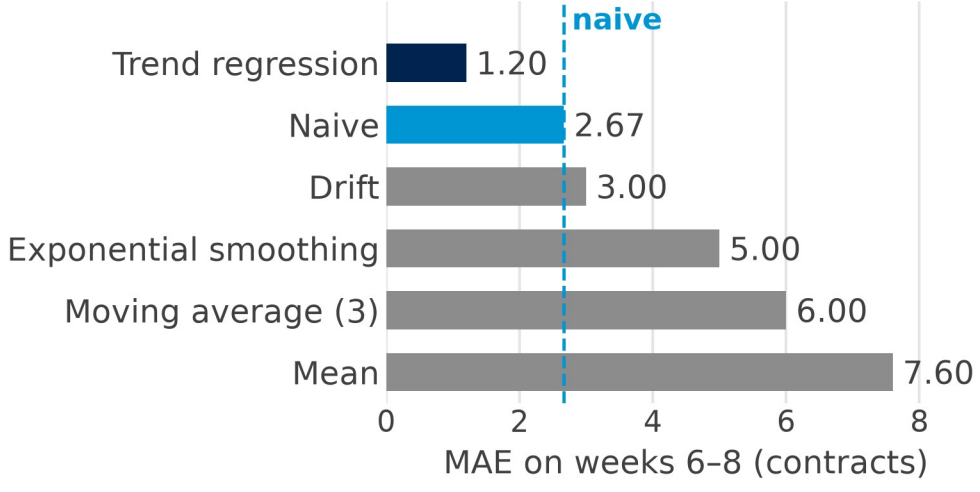
**2.94** **RMSE:** √8.67, big errors count more

**5.0 %** **MAPE:** mean of |error| / actual

**Error = actual − forecast:** a plus means the forecast was too low.

Invented round numbers (hand_examples.csv)

# The ranking: beat the simplest benchmark




**1.20** **trend regression:** the only method that beats naive (2.67)

**4.00** **chocolate shop, year 3 (MAE, k EUR):** seasonal naive 4.00 vs naive 8.00, so seasonal data need a seasonal benchmark

Forecasts for weeks 6–8, fitted on weeks 1–5: trend regression 37.8 + 2.2 × week = 51, 53.2, 55.4; naive 50; drift 52.5, 55, 57.5; exponential smoothing 47; moving average (3) 46; mean 44.4. RMSE and MAPE give the same order.

**Simplest benchmark first:** naive, or seasonal naive for seasonal data; keep what beats it.

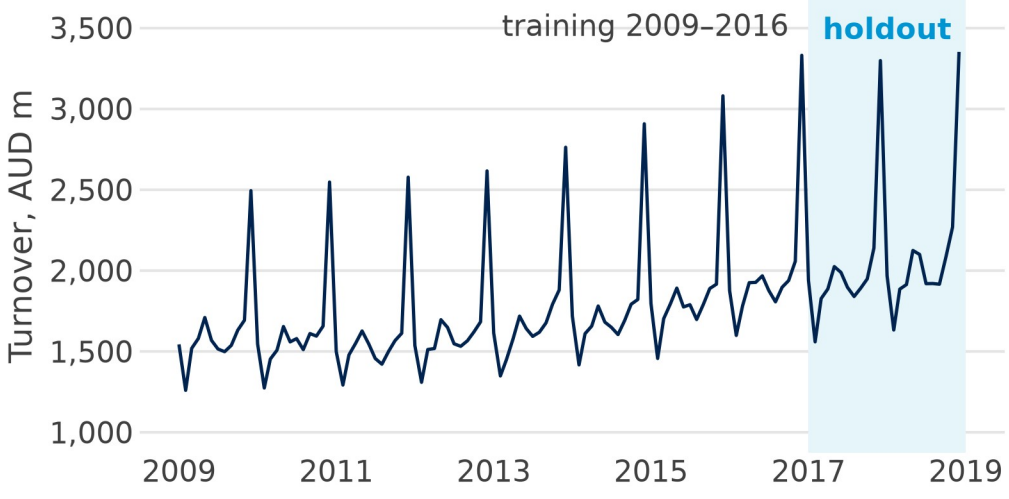
Invented round numbers (hand_examples.csv)

# Real data: clothing sales in Australia

Part 4 of 6 · trend, season, holdout, 2019

# The data: monthly clothing sales, 2009–2018

**The task:** a fashion retailer plans stock for 2019 from the market's monthly sales.




**Source:** Australian Bureau of Statistics (ABS), Retail Trade, open licence (CC BY 4.0)

**Unit:** turnover, AUD million a month, not seasonally adjusted

**Period:** Jan 2009 to Dec 2018, 120 months: 96 to fit, 24 holdout

**File:** `data/forecasting/aus_clothing_retail.csv` one row per month: month, value

Source: ABS, Retail Trade, Australia (8501.0), Table 11, clothing, footwear and personal accessories; CC BY 4.0

# The two real data sets: source and access

| | Clothing | Cafés |
|---|---|---|
| **Industry** | clothing, footwear, accessories | cafés, restaurants, takeaway |
| **ABS series ID** | A3349787F | A3349870V |
| **Months** | April 1982 – April 2019 (445) | April 1982 – April 2019 (445) |
| **Used in** | the deck, forecasting.qmd | the in-class exercise |
| **File** | aus_clothing_retail.csv | aus_cafes_retail.csv |

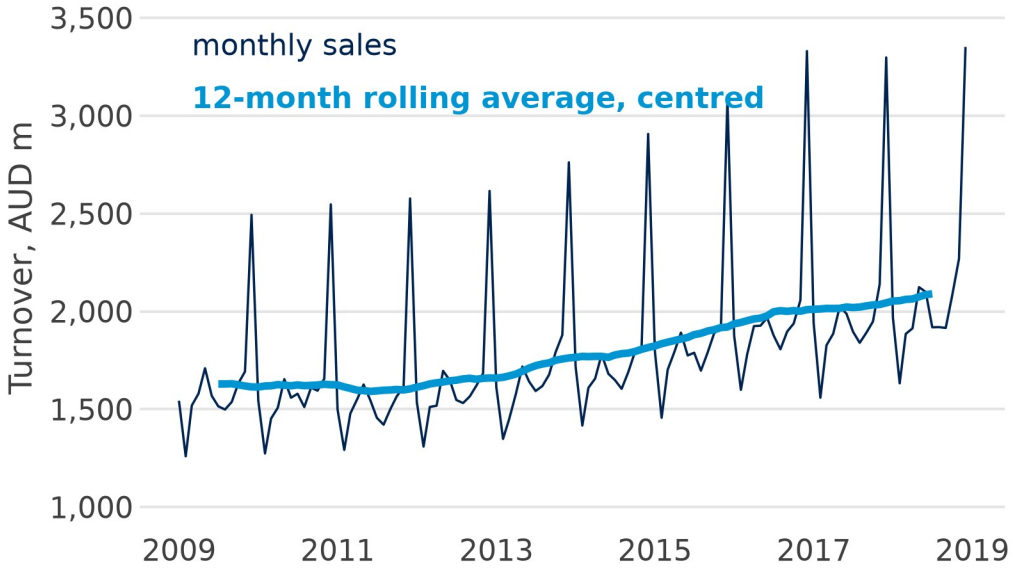
**Course files:** data/forecasting/ in the repository; on GitHub: Code → Download ZIP

**Original:** abs.gov.au → Retail Trade, Australia (cat. 8501.0), time-series Table 11

**Textbook:** the same data as aus_retail in FPP3 (otexts.com/fpp3)

Source: ABS, Retail Trade, Australia, cat. no. 8501.0, April 2019, Table 11; © Commonwealth of Australia, CC BY 4.0

# A 12-month rolling average shows the trend




**1,629** **Jul 2009:** the average month of 2009 (19,546 / 12)

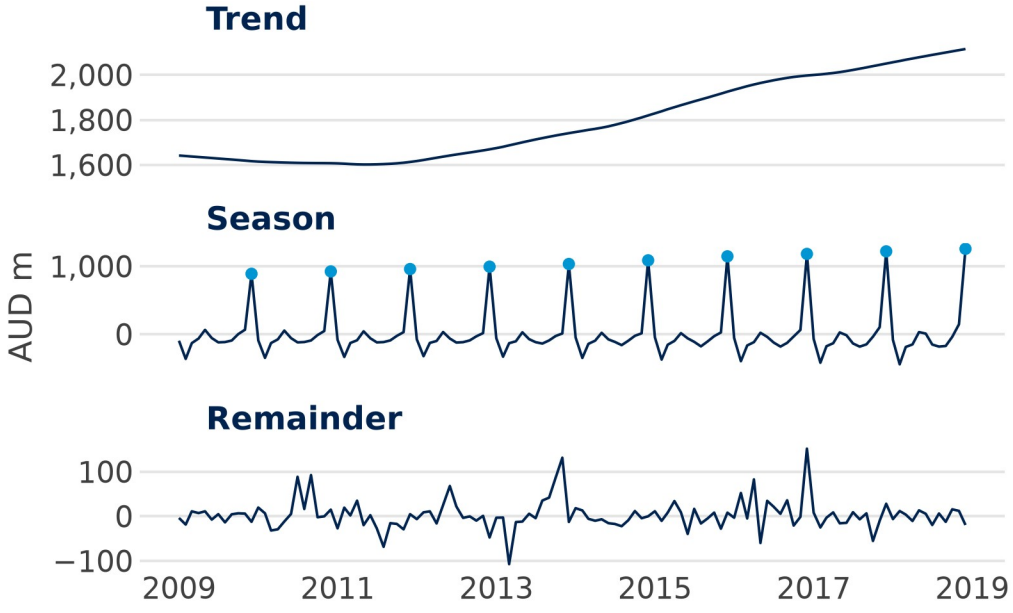
**2,090** **Jul 2018:** the average month of 2018 (25,077 / 12): +28 %

**1,591** **lowest, Jun 2011:** flat to 2012, then growth: the cycle

**Twelve months around each month cancel the season:** STL (next slide) is a smarter version.

Source: ABS Retail Trade (8501.0), clothing; CC BY 4.0

# Trend, season and cycle in the real series




**× 1.59** — **December** vs an average month; February × 0.80

**+28 %** — **2009 to 2018;** flat to 2012 (a cycle), then about +4 % a year

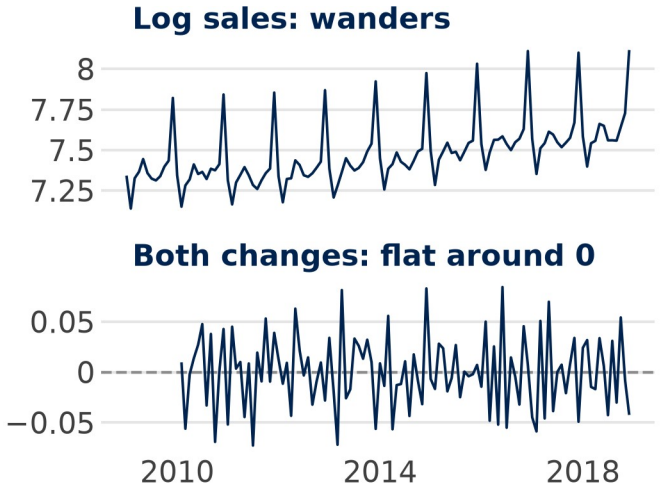
**1,257** — **December swing** in 2018, AUD m (2009: 890): it grows with the level

**Strong trend and season:** a good method must model both.

Source: ABS Retail Trade (8501.0), clothing; CC BY 4.0

# Is it stationary? The ADF test

| Series | ADF p | Stationary? |
|---|---|---|
| **Sales (levels)** | 0.992 | no |
| **Log sales** | 0.987 | no |
| **Change on last month** | 0.011 | yes (season stays) |
| **Change on last year** | 0.003 | yes |
| **Both changes, log sales** | **< 0.001** | **yes** |




**ADF in plain words:** does it wander without a fixed mean? Small p (below 0.05): no, stationary.

**Difference until it is stationary:** SARIMA does it for us (next slide).

Source: ABS Retail Trade (8501.0), clothing; CC BY 4.0

# Two models that learn trend and season

## Holt-Winters / ETS

- Smooths level, trend and season (multiplicative)
- Each has its own weight α, β, γ (slide 15)
- ETS: the same idea as a statistical model, so it also gives bands

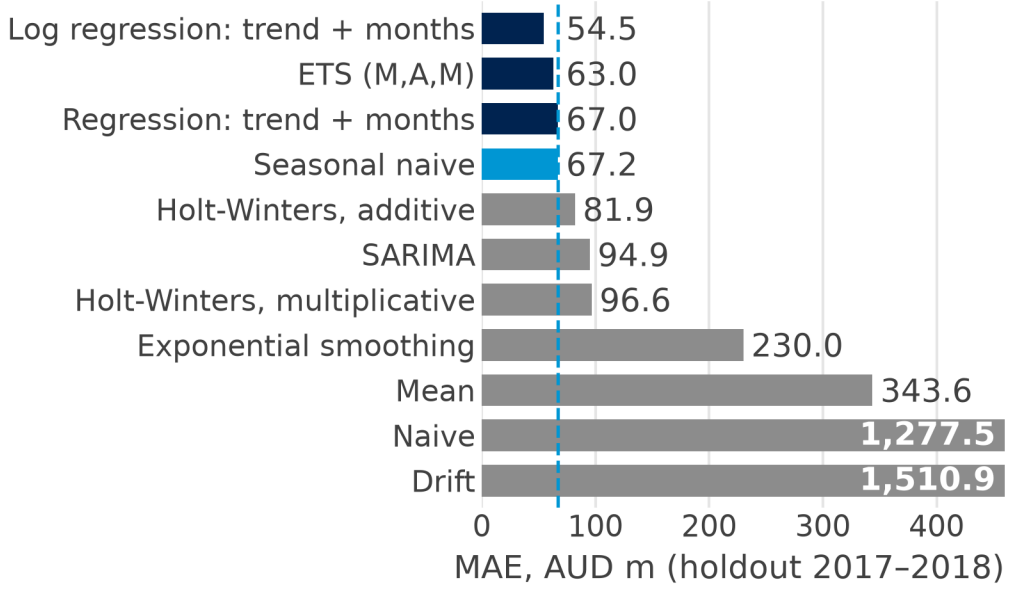
## SARIMA

- Differences remove trend and season
- Then it models the memory: how much of last month's and last year's surprise carries on
- Ours: SARIMA$(0,1,1)(0,1,1)_{12}$ on log sales

log sales → change on last month → change on last year → memory of surprises

**Both learn trend and season from the data;** the holdout decides which forecasts better.

# Holdout 2017–2018: beat the seasonal naive




**3** of the other 14 methods beat the **seasonal naive** (67.2)

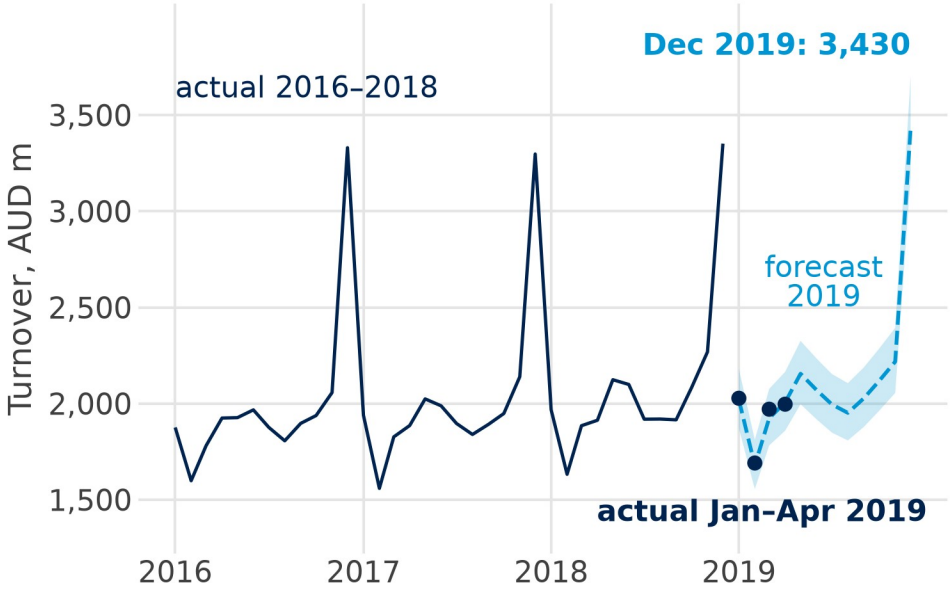
**54.5** **best:** log regression, trend + months (MAPE 2.5 %)

**3,331** **December trap:** naive repeats Dec 2016 for 24 months (MAE 1,277.5)

**On seasonal data the seasonal naive is the benchmark:** only three methods beat it.

Source: ABS Retail Trade (8501.0), clothing; CC BY 4.0; holdout 2017–2018

# The 2019 forecast with a 95 % band




**3,430** — **December 2019;** 95 % band 3,178 to 3,702: should hold the actual in 95 of 100 cases

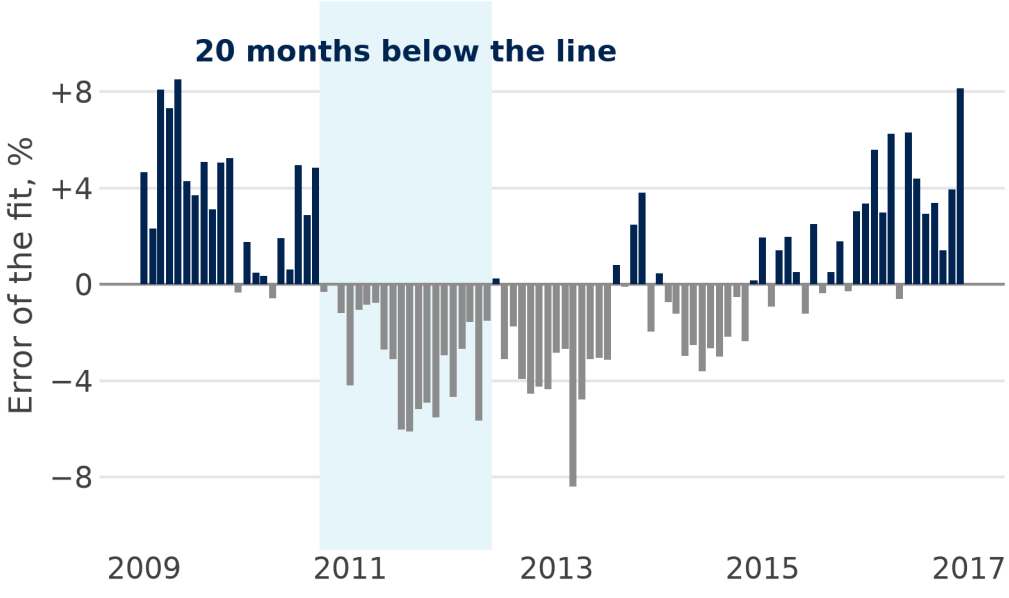
**1.0 %** — **error Jan–Apr 2019** (MAPE); all 4 inside the band

**25,595** — **forecast for 2019** (2018: 25,077)

**Refit the winner on all 120 months,** then report the forecast with its band.

Source: ABS Retail Trade (8501.0), clothing; CC BY 4.0

# Autocorrelation: errors that stick together




**0.72** **autocorrelation:** this month's error is like last month's (0 = none)

**0.49** **Durbin–Watson:** 2 = no autocorrelation

**524** **December band width,** AUD m (regression); SARIMA's widens from 209 in Jan to 589

**Errors that stick together:** the regression band does not widen with the horizon; for later months use the SARIMA band.

Source: ABS Retail Trade (8501.0), clothing; CC BY 4.0; training months 2009–2016

# Forecasting in Python




Part 5 of 6 · a code ladder, one step at a time

# The packages and functions

| Task | Package | Function |
|---|---|---|
| **Load and look** | polars | `pl.read_csv(); filter(); head()` |
| **Plot the series** | plotnine | `ggplot() + geom_line()` |
| **Forecast and MAE** | numpy | `np.repeat(); np.resize(); np.mean()` |
| **Smooth** | polars | `rolling_mean(12, center=True)` |
| **Exponential smoothing** | statsmodels | `tsa.ExponentialSmoothing()` |
| **Regression** | statsmodels | `smf.ols("np.log(value) ~ t + C(m)")` |
| **SARIMA** | statsmodels | `tsa.SARIMAX()` |
| **Forecast with band** | statsmodels | `get_prediction(new).summary_frame()` |

# Step 1: load and look



**forecasting.qmd · Step 1**

```
import polars as pl
from plotnine import aes, geom_line, ggplot

# load the monthly sales and keep 2009-2018
file = "../../data/forecasting/aus_clothing_retail.csv"
df = pl.read_csv(file, try_parse_dates=True)
df = df.filter(
    pl.col("month").dt.year().is_between(2009, 2018))
print(df.head())

# plot the series
ggplot(df, aes("month", "value")) + geom_line()
```

| df.head() | value |
|---|---|
| Jan 2009 | 1,543.2 |
| Feb 2009 | 1,259.1 |
| Mar 2009 | 1,518.6 |

**Look first:** sales grow, and the December peak grows with them (slide 25).

# Step 2: split: fit and holdout



**forecasting.qmd · Step 2**

```
# the sales as numbers, in time order
y = df["value"].to_numpy()

# fit on the first 96 months, hold out the last 24
train, test = y[:-24], y[-24:]
print(len(train), len(test))
```

| Part | Months | Period |
|---|---|---|
| train | 96 | 2009–2016 |
| test | 24 | 2017–2018 |



**Test on the future:** fit on 2009–2016, forecast 2017–2018; never split at random.

# Step 3: naive and seasonal naive



**forecasting.qmd · Step 3**

```
import numpy as np

# naive: repeat the last training month 24 times
naive = np.repeat(train[-1], 24)

# seasonal naive: repeat the last training year twice
snaive = np.resize(train[-12:], 24)
print(naive[:12])
print(snaive[:12])
```

| 2017 | naive | snaive |
|---|---|---|
| Jan | 3,330.5 | 1,875.7 |
| Jun | 3,330.5 | 1,967.1 |
| Dec | 3,330.5 | 3,330.5 |

**Naive forecasts a Christmas every month;** the seasonal naive repeats 2016.

# Step 4: how good? The MAE



**forecasting.qmd · Step 4**

```
# MAE: the mean absolute error on the holdout
mae_snaive = np.mean(np.abs(test - snaive))
mae_naive = np.mean(np.abs(test - naive))
print(mae_snaive, mae_naive)
```

| Method | MAE |
|---|---|
| snaive | 67.2 |
| naive | 1,277.5 |



**The December trap:** naive misses by 1,277.5 a month; beat the seasonal naive.

# Step 5: smooth: a rolling average



**forecasting.qmd · Step 5**

```
# 12-month rolling average, centred: the trend
smooth = df.with_columns(
    trend=pl.col("value").rolling_mean(12, center=True))
print(smooth.drop_nulls())

# the sales and the trend in one chart
chart = ggplot(smooth, aes("month", "value")) + geom_line()
chart + geom_line(aes(y="trend"), color="blue")
```

| trend | AUD m |
|---|---|
| Jul 2009 | 1,628.8 |
| Jul 2018 | 2,089.8 |



**The season cancels out:** the trend is flat to 2012, then grows (+28 %).

# Step 6: exponential smoothing



**forecasting.qmd · Step 6**

```
import statsmodels.tsa.api as tsa

# Holt-Winters: a trend and a season that grows
hw = tsa.ExponentialSmoothing(
    train, trend="add", seasonal="mul",
    seasonal_periods=12,
    initialization_method="estimated").fit()
hw_fc = hw.forecast(24)
mae_hw = np.mean(np.abs(test - hw_fc))
print(mae_hw)
```

| Method | MAE |
|---|---|
| **Holt-Winters** | 96.6 |
| **snaive** | 67.2 |

**Smarter is not always better:** Holt-Winters loses to the seasonal naive.

# Step 7: regression: trend and months



**forecasting.qmd · Step 7**

```
import statsmodels.formula.api as smf

# t counts the months 1-120, m is the month 1-12
d = df.with_columns(
    t=pl.int_range(1, pl.len() + 1),
    m=pl.col("month").dt.month()).to_pandas()
# log sales on trend + month dummies, first 96 months
reg = smf.ols("np.log(value) ~ t + C(m)",
              data=d[:-24]).fit()
reg_fc = np.exp(reg.predict(d[-24:]))
mae_reg = np.mean(np.abs(test - reg_fc))
print(mae_reg)
```

| Method | MAE |
|---|---|
| regression | 54.5 |
| snaive | 67.2 |

**The first winner:** trend + months on log sales beats the seasonal naive.

# Step 8: SARIMA



**forecasting.qmd · Step 8**

```
# SARIMA on log sales: a change on last month
# and on last year, and the memory of surprises
sar = tsa.SARIMAX(
    np.log(train), order=(0, 1, 1),
    seasonal_order=(0, 1, 1, 12)).fit(disp=0)
sar_fc = np.exp(sar.forecast(24))
mae_sar = np.mean(np.abs(test - sar_fc))
print(mae_sar)
```

| Method | MAE |
|---|---|
| **SARIMA** | 94.9 |
| **snaive** | 67.2 |

**Test, do not assume:** SARIMA also loses to the seasonal naive here.

# Step 9: compare the four



**forecasting.qmd · Step 9**

```
# the four MAEs side by side
pl.DataFrame({
    "method": ["snaive", "hw", "reg", "sarima"],
    "MAE": [mae_snaive, mae_hw, mae_reg, mae_sar]})
```

| method | MAE |
|---|---|
| snaive | 67.2 |
| hw | 96.6 |
| reg | 54.5 |
| sarima | 94.9 |

**The regression wins:** the same ranking as slide 31.

# Step 10: forecast 2019



**forecasting.qmd · Step 10**

```
# refit the winner on all 120 months
reg_all = smf.ols("np.log(value) ~ t + C(m)",
                  data=d).fit()

# 2019: t = 121 to 132, m = 1 to 12
new = pl.DataFrame({"t": range(121, 133),
                    "m": range(1, 13)}).to_pandas()

# the forecast and its 95 % band, back from logs
band = reg_all.get_prediction(new).summary_frame()
np.exp(band[["mean", "obs_ci_lower", "obs_ci_upper"]])
```

| 2019 | mean | low | high |
|---|---|---|---|
| Jan | 2,020.4 | 1,871.9 | 2,180.6 |
| Dec | 3,430.2 | 3,178.2 | 3,702.2 |

**Plan December with 3,430 AUD m:** the 95 % band runs from 3,178 to 3,702.

# In-class exercise




Part 6 of 6 · quiz, exercise, Posit Assistant

# Online quiz: forecasting in five questions

**1 Sales: 50, 54, 58. The drift (same change) forecast for the next period is …**
A 58 B 62 C 54

**2 Exponential smoothing with α 0.5: level 40, new value 50. The new level is …**
A 45 B 50 C 42

**3 Holdout errors: −2, +2, +4. The MAE is …**
A 1.33 B 4.00 C 2.67

**4 Monthly sales with a December peak. The benchmark to beat is …**
A naive (last month) B seasonal naive (same month last year) C the mean

**5 ADF test on sales levels: p 0.99. The series is …**
A stationary B white noise C not stationary: difference it

**3 minutes alone,** then compare with your neighbour.

# Answers: in class

# Your turn: forecast café sales in Python

| Pairs: 35 minutes | Plenary: 5 minutes | Posit Assistant: 10 minutes |
|---|---|---|

1. **By hand:** café bookings: five forecasts for week 7, MAE on weeks 5–6 (Steps 3, 4; Extra A)
2. **Look:** plot 2009–2018; trend, season, ADF (Step 1; Extra C)
3. **Test:** five methods on the 2017–2018 holdout: MAE, RMSE, MAPE (Steps 3, 6–9)
4. **Forecast:** 2019 with the winner and SARIMA, 95 % bands; check Jan–Apr (Step 10; Extra D)

**Deliverable:** one slide per pair: the 2019 forecast and two sentences.

**File:** `sessions/01-foundations/forecasting_exercise.qmd` (reuse: forecasting.qmd)

# Then: the same tasks with Posit Assistant

**How:** code first, then open Posit Assistant, type @forecasting_exercise.qmd, one task per prompt

### Prompt for task 1

```
@forecasting_exercise.qmd Task 1. Café
bookings 20, 24, 22, 26, 27, 31: naive,
drift, mean, moving average (3) and SES
(α 0.5) for week 7, step by step.
```

### Prompt for task 2

```
Task 2. Plot aus_cafes_retail.csv for
2009-2018 with plotnine. Describe trend
and season; run the ADF test on levels
and on the first difference.
```

### Prompt for task 3

```
Task 3. Hold out 2017-2018. Compare
seasonal naive, Holt-Winters, SARIMA and
two regressions by MAE, RMSE and MAPE
in a great_tables table.
```

### Prompt for task 4

```
Task 4. Forecast 2019 with the best
method and with SARIMA, 95 % bands.
Which band would you trust for stock
planning, and why?
```

**Check every number against your own code:** you own the result.

# Open data for forecasting

## ABS Retail Trade

Monthly turnover in Australia since 1982, by state and industry

**Licence:** CC BY 4.0

## Eurostat

Monthly retail trade and tourism nights by EU country

**Licence:** free reuse, cite the source

## M5 (Walmart)

Daily unit sales of 3,049 products in 10 US stores, 2011–2016

**Licence:** competition data: check the terms

**The licence decides:** commit only data you may share; link the rest.

**Links and loading code:** `forecasting.qmd` (appendix)

Sources: ABS, Eurostat and M5 dataset pages (links in forecasting.qmd)

# Key takeaways

1. **Plot and smooth first:** trend, season, cycle; additive or multiplicative; stationary or not?
2. **Start simple:** naive, drift, averages, smoothing and a trend line: all on a calculator
3. **Test on the future:** hold out the last periods; compare MAE, RMSE and MAPE
4. **Beat the simplest benchmark:** on seasonal data, the seasonal naive
5. **Report the band, check the errors:** autocorrelated errors make a band too narrow

**Next:** run the ten steps of sessions/01-foundations/forecasting.qmd, then the exercise.

# References


## Forecasting methods

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**Hyndman, R. J. & Koehler, A. B. (2006).** Another look at measures of forecast accuracy. International Journal of Forecasting, 22(4), 679–688.

**Cleveland, R. B. et al. (1990).** STL: A seasonal-trend decomposition procedure based on loess. Journal of Official Statistics, 6(1), 3–73.

## Competitions, data and software

**Makridakis, S., Spiliotis, E. & Assimakopoulos, V. (2020).** The M4 Competition: 100,000 time series and 61 forecasting methods. International Journal of Forecasting, 36(1), 54–74.

**Makridakis, S. et al. (2022).** M5 accuracy competition: Results, findings, and conclusions. International Journal of Forecasting, 38(4), 1346–1364.

**Australian Bureau of Statistics (2019).** Retail Trade, Australia, April 2019, cat. no. 8501.0, Table 11. CC BY 4.0.

**Seabold, S. & Perktold, J. (2010).** statsmodels: Econometric and statistical modeling with Python. Proceedings of the 9th Python in Science Conference.


**Free textbook:** `otexts.com/fpp3` · Python edition: `otexts.com/fpppy`

# Thank you. Questions and discussion

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